Mean, Mode & Median of ungrouped data

Measure of Central Tendency

Measures of Central Tendency

A measure of central tendency is a descriptive statistic that describes the average, or typical value of a set of scores Since such typical values tend to lie centrally within a set of data arranged according to magnitude, averages are also called measures of central tendency.

- There are five common measures of central tendency:
  - the mode
  - the median
  - the mean
  - Geometric mean
  - Harmonic mean

The Median

The median is the middle value in distribution when the values are arranged in ascending or descending order

60, 58, 57, 55, 56, 54, 54, 60, 57, 54, 58

Arranged in ascending

54, 54, 54, 55, 56, **57**, 57, 58, 58, 60, 60

The median is the middle value, which is 57

When the distribution has an even number of observations, the median value is the mean of the two middle values

52, 54, 54, 54, 55, **56, 57**, 57, 58, 58, 60, 60

Two middle values are 56 and 57, therefore the median equals 56.5

The Median

- It is the score in the middle; half of the scores are larger than the median and half of the scores are smaller than the median

- The median divides the distribution in half (there are 50% of observations on either side of the median value).

- In a distribution with an odd number of observations, the median value is the middle value.

- When the distribution has an even number of observations, the median value is the mean of the two middle values.

Alternative way to find Median

Sort the data from highest to lowest

If the total number of observations given is odd, then the formula to calculate the median is:

$$\text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{Observation}$$

If the total number of observation is even, then the median formula is:

$$\textbf{Median} = \frac{\left(\frac{n}{2}\right)^{th} \textbf{observation} + \left(\frac{\mathbf{n}}{\mathbf{2}} + \mathbf{1}\right)^{\text{th}} \textbf{observation}}{\mathbf{2}}$$

Example: The age of the members of a weekend poker team has been listed below. Find the median of the above set.
42, 40, 50, 60, 35, 58, 32

Step 1: Arrange the data items in ascending order.

Original set: 42, 40, 50, 60, 35, 58, 32

Ordered Set: 32, 35, 40, 42, 50, 58, 60

Step 2: Count the number of observations. If the number of observations is odd, then we will use the following formula:

Median = $\left(\frac{n+1}{2}\right)^{th}$ term

Step 3: Calculate the median using the formula.

$$\text{median} = \left(\frac{7+1}{2}\right)^{th} \text{term} = 4^{\text{th}} \text{term} = 42$$

Median of even observation

52, 54, 54, 54, 55, 56, 57, 57, 58, 58, 60, 60

$$\textbf{Median} = \frac{\left(\frac{n}{2}\right)^{th} \textbf{ observation} + \left(\frac{n}{2} + 1\right)^{th} \textbf{ observation}}{2}$$

$$= \frac{\left(\frac{12}{2}\right)^{th} term + \left(\frac{12}{2} + 1\right)^{th} term}{2}$$

$$= 6.5^{th} \text{ term}$$

$$= \text{taking average of } 6^{\text{th}} \ \& \ 7^{\text{th}} \text{ term}$$

$$= \frac{56 + 57}{2}$$

$$= 56.5$$

Median for grouped data

For a set of **grouped data**, we can follow the following steps to find the median: When the data is continuous and in the form of a frequency distribution, the median is calculated through the following sequence of steps.

•Step 1: Find the total number of observations $n = \sum f$

•Step 2: Define the class size(h), and divide the data into different classes.

•Step 3: Calculate the cumulative frequency of each class.

•Step 4: Identify the class in which the median falls. (Median Class is the class where n/2 lies.)

•Step 5: Find the lower limit of the median class(l), and the cumulative frequency(c).

•Step 6: Apply the formula

$$\text{Median} = l + \left[\frac{\frac{n}{2} - c}{f} \right] \times h$$

where,

- l = lower limit of median class
- c = cumulative frequency of the class preceding the median class
- f = frequency of the median class
- h = class size

Example: Find the median marks for the following distribution:

Classes	0-10	10-20	20-30	30-40	40-50
Frequency	2	12	22	8	6

We need to calculate the cumulative frequencies to find the median.

Calculation table:

Classes	Number of students	Cumulative frequency
0-10	2	2
10-20	12	2 + 12 = 14
20-30	22	14 + 22 = 36
30-40	8	36 + 8 = 44
40-50	6	44 + 6 = 50

$$n = \sum f = 50 \text{ (last answer of cumulative frequency)}$$

$$\frac{n}{2} = \frac{50}{2} = 25$$

As 25 lies in third class so median class is **20-30**

$l = 20, f = 22, c = 14, h = 10$

Using Median formula:

$$\text{Median} = l + \left[\frac{\frac{n}{2} - c}{f}\right] \times h$$

$$= 20 + (25 - 14)/22 \times 10$$

$$= 20 + (11/22) \times 10$$

$$= 20 + 5 = 25$$

$$\therefore \text{Median} = 25$$

Mean for ungrouped data

The mean (or arithmetic mean) of n observations (variates) x_1, x_2, x_3, x_4, ..., x_n is given by

Mean = $\dfrac{x_1+x_2+x_3+x_4+....+x_n}{n}$

In words, mean = $\dfrac{\textbf{Sum of the Variables}}{\textbf{Total Number of Variates}}$

Symbolically, A = $\dfrac{\sum x_i}{n}$; i = 1, 2, 3, 4, ..., n.

Example

Sachin Tendulkar scores the following runs in six innings of a series.

45, 2, 78, 20, 116, 55.

Find the mean of the runs scored by the batsman in the series

Solution:

Here, the observations are $x_1 = 45$, $x_2 = 2$, $x_3 = 78$, $x_4 = 20$, $x_5 = 116$, $x_6 = 55$.

Therefore, the required mean $= \dfrac{x_1+x_2+x_3+x_4+x_5+x_6}{6}$

$$= \frac{45+2+78+20+116+55}{6}$$

$$= \frac{316}{6}$$

$$= 52.7.$$

Therefore, the mean of the runs scored by Sachin Tendulkar in the series is 52.7.

Note

The mean of the runs scored by the batsman in six innings indicates the batsman's form, and one can expect the batsman to score about 53 runs in his next outing. However, it may so happen that the batsman scores a duck (0) or a century (100) the next time he bats.

Mean of Grouped Data

If $x_1, x_2, x_3, \ldots x_n$ are observations having respective frequencies $f_1, f_2, \ldots f_n$.

This means that x_1 occurs f_1 times, x_2 occurred f_2 times and so on. Then,

Sum of the values of observations $= x_1 f_1 + x_2 f_2 + \cdots + x_n f_n$

and sum of frequencies $= f_1 + f_2 + \cdots + f_n$

$$\overline{X} = \frac{f_1 x_1 + f_2 x_2 + \cdots + f_n x_n}{f_1 + f_2 + \cdots + f_n}$$

$$= \frac{\sum_{i=1}^{n} x_i f_i}{\sum_{i=1}^{n} f_i}$$

Methods for Calculating Mean

Step 1: For each class, find the class mark x_i as

$$x = \frac{1}{2}\left(\text{upper limit} + \text{lower limt}\right)$$

Step 2: $\text{calculate } f_i x_i \text{ for each } i$

Step 3: Calculate mean by

$$\overline{x} = \frac{\sum fx}{\sum f}$$

Example 1: Find the mean of the following distribution:

x	**4**	**6**	**9**	**10**	**15**
f	**5**	**10**	**10**	**7**	**8**

Solution:

Calculation table for arithmetic mean:

x_i	f_i	$x_i f_i$
4	**5**	**20**
6	**10**	**60**
9	**10**	**90**
10	**7**	**70**
15	**8**	**120**
	$\sum f_i = 40$	$\sum x_i f_i = 360$

$$\text{Mean}, \bar{x} = \frac{\sum x_i f_i}{\sum f_i}$$

$= 360/40$

$= 9$

Thus, Mean = 9

Example 2

Number of patients	Number of days visiting hospital
0-10	2
10-20	6
20-30	9
30-40	7
40-50	4
50-60	2

In this case, we find the classmark (also called as mid-point of a class) for each class.

Note: Class mark = (lower limit + upper limit)/2

Let $x_1, x_2, x_3 \ldots\ldots x_n$ be the class marks of the respective classes.

Hence, we get the following table:

Class mark (x_i)	**frequency (f_i)**	**$x_i f_i$**
5	2	10
15	**6**	90
25	**9**	225
35	**7**	245
45	**4**	180
55	**2**	110
Total	$\sum f_i = 30$	$\sum f_i x_i = 860$

$$\text{Mean, } \bar{x} = \frac{\sum x_i f_i}{\sum f_i}$$

$= 860/30$

$= 28.67$

$\bar{x} = \mathbf{28.67}$

Mode for ungrouped data

The value or number which occurs maximum time in the data is regarded as mode. The ungrouped data contains discrete values. To find the mode that is the most frequent value in the data set, we need to arrange the data in ascending order or descending order and then find the frequencies of each value. The value which has maximum frequency will be the mode of ungrouped data

Example 1: Find the mode of the following marks obtained by 25 students in a mathematics test out of 50.
34, 46, 45, 39, 43, 22, 27, 37, 46, 35, 34, 39, 40, 30, 30, 41, 37, 46, 39, 29, 34, 39, 35, 43, 30

Solution:
The ascending order of the data:
22, 27, 29, 30, 30, 30, 34, 34, 34, 35, 35, 37, 37, 39, 39, 39, 39, 40, 41, 43, 43, 45, 46, 46, 46

The mode of the given data can be obtained by making the frequency table and choosing the highest frequency. Such as:

Observation	22	27	29	30	34	35	37	39	40	41	43	45	46
Frequency	1	1	1	3	3	2	2	4	1	1	2	1	3

Here, the highest frequency is 4.
Therefore, the mode is 39.

How to Calculate Mode Step by Step?

Step 1. Find the maximum class frequency.

Step 2. Find the class corresponding to this frequency. It is called the modal class.

Step 3 . Then, we will find the value of h, which is the size of the class interval, and can be calculated by using the formula, upper limit – lower limit.

Step 4. Calculate mode using the formula.

$$\text{Mode} = l + \left[\frac{f_m - f_1}{2f_m - f_1 - f_2}\right] \times h$$

where,

- l = lower limit of modal class,
- f_m = frequency of modal class,
- f_1 = frequency of class preceding modal class,
- f_2 = frequency of class succeeding modal class,
- h = class width

Example: Find the mode of the given data:

Marks Obtained	0-20	20-40	40-60	60-80	80-100
Number of students	5	10	12	6	3

Marks Obtained	0-20	20-40	40-60	60-80	80-100
Number of students	5	10	12	6	3

Solution:

The highest frequency = 12, so the modal class is 40-60.

l = lower limit of modal class = 40

f_m = frequency of modal class = 12

f_1 = frequency of class preceding modal class = 10

f_2 = frequency of class succeeding modal class = 6

h = class width = 20

$$\text{Mode} = l + \left[\frac{f_m - f_1}{2f_m - f_1 - f_2}\right] \times h$$

$$= 40 + \left[\frac{12 - 10}{2 \times 12 - 10 - 6}\right] \times 20$$

$$= 40 + (2/8) \times 20$$

$$= 45$$

Relation between Mean, Median and Mode

The three measures of central values i.e. mean, median and mode are closely connected by the following relations (called an **empirical relationship**).

2Mean + Mode = 3Median

For instance, if we are asked to calculate the mean, median, and mode of continuous grouped data, then we can calculate mean and median using the formulas as discussed in the previous sections and then find mode using the empirical relation.

For example, we have data whose mode = 65 and median = 61.6.

Then, we can find the mean using the above mean, median, and mode relation.

$$2\text{Mean} + \text{Mode} = 3\ \text{Median}$$

$$\therefore 2\text{Mean} = 3 \times 61.6 - 65$$

$$\therefore 2\text{Mean} = 119.8$$

$$\Rightarrow \text{Mean} = 119.8/2$$

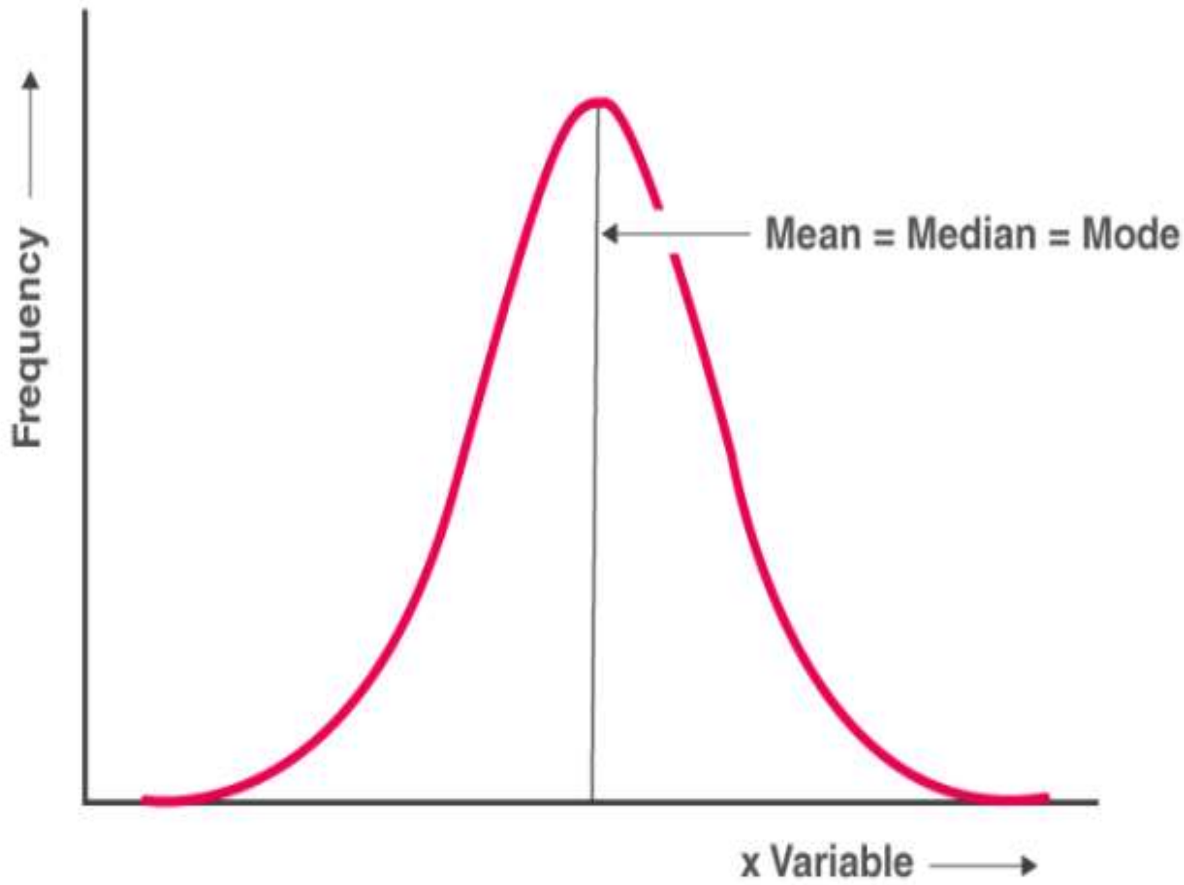
$$\Rightarrow \text{Mean} = 59.9$$

Mean Median Mode Relation with Frequency Distribution

Frequency Distribution with Symmetrical Frequency Curve

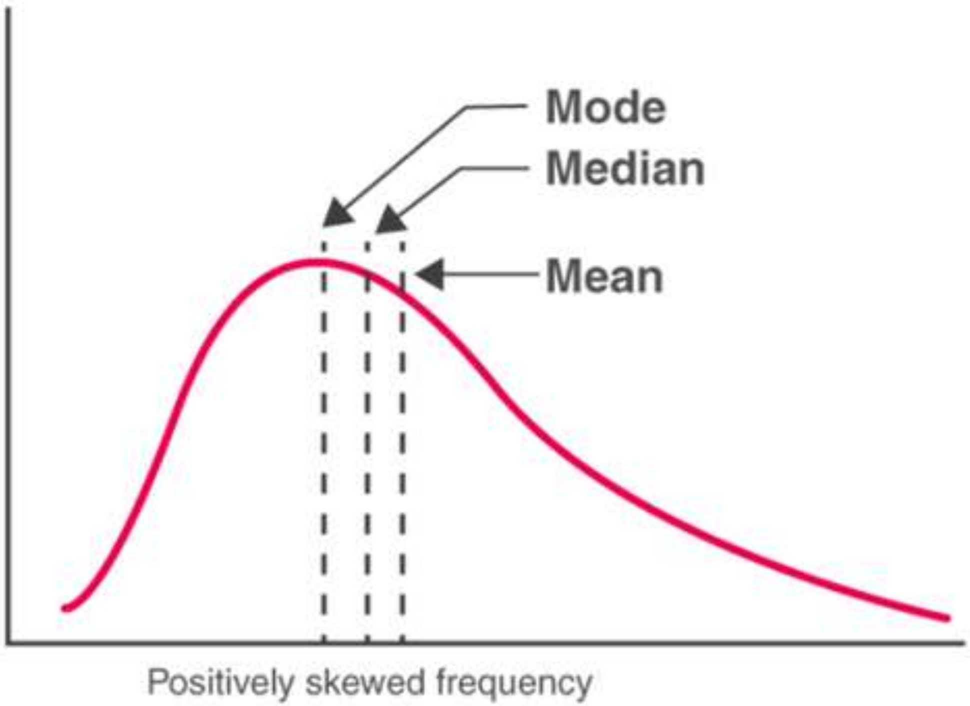
If a frequency distribution graph has a symmetrical frequency curve, then mean, median and mode will be equal.

Mean = Median = Mode


Frequency
Mean = Median = Mode
x Variable

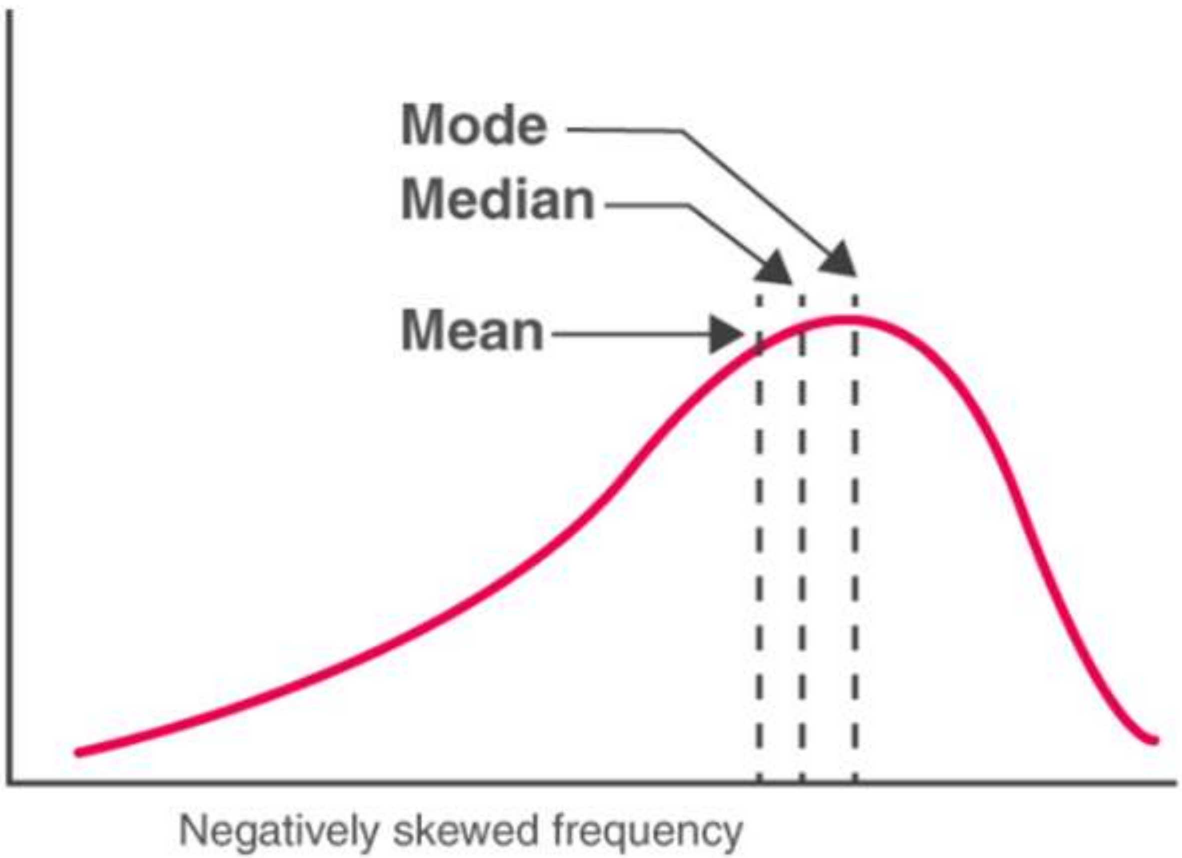
Positively Skewed Frequency Distribution

In case of a positively skewed frequency distribution, the mean is always greater than median and the median is always greater than the mode



For Negatively Skewed Frequency Distribution

In case of a negatively skewed frequency distribution, the mean is always lesser than median and the median is always lesser than the mode



Mean < Median < Mode

Geometric and harmonic mean

G.M for Ungrouped data

If $x_1, x_2, \ldots, x_n$ are observations

$$\text{GM} = \text{Antilog}\, \frac{\sum \log x_i}{n}$$

For grouped data

$$\text{GM} = \text{Antilog}\left[\frac{\sum f \log x_i}{n}\right]$$

H.M for Ungrouped data

$$\text{H.M} = \frac{n}{\sum_{i=n}^{n}\left(\frac{1}{x_i}\right)}$$

H.M for grouped data

$$\text{H.M} = \frac{n}{\sum_{i=n}^{n} f\left(\frac{1}{x_i}\right)}$$

Calculate the geometric mean of the annual percentage growth rate of profits in business corporate from the year 2000 to 2005 is given below

50, 72, 54, 82, 93

x_i	50	72	54	82	93	**Total**
$\log x_i$	1.6990	1.8573	1.7324	1.9138	1.9685	**9.1710**

$$\text{G.M.} = \text{Antilog} \frac{\sum_{i=1}^{n} \log x_i}{n}$$

$$= \text{Antilog} \frac{9.1710}{5}$$

$$= \text{Antilog } 1.8342$$

$$\text{G. M.} = 68.26$$

Example

Find geometric mean and harmonic mean of given data.

Class	Frequency
2-4	3
4-6	4
6-8	2
8-10	1

Class	Mid values (x)	frequency	flog(x)	$\frac{f}{x}$
2-4	3	3	1.4314	1
4-6	5	4	2.7959	0.8
6-8	7	2	1.6902	0.2857
8-10	9	1	0.9542	0.1111
		$n = 10$	$\sum flog = 6.8717$	$\sum \frac{f}{x} = 2.1968$

$$G.M = Antilog\left(\sum \frac{flog}{n}\right) \qquad G.M = Antilog\left(\frac{6.8717}{10}\right) = 4.866$$

$$H.M = \frac{n}{\sum\left(\frac{f}{x}\right)} = \frac{10}{2.1968} = 4.552$$

Example

Find geometric mean and harmonic mean of given data.

x	Frequency
10	3
11	12
12	18
13	12
14	3

x	frequency	f log(x)	$\frac{f}{x}$
10	3	3	0.3
11	12	12.4967	1.0909
12	18	19.4253	1.5
13	12	13.3673	09231
14	3	3.4384	0.2143
	n = 48	$\sum flog = 51.7277$	$\sum \frac{f}{x} = 4.0283$

$$G.M = Antilog\left(\sum \frac{flog}{n}\right)$$

$$G.M = Antilog\left(\frac{51.7277}{48}\right) = 11.958$$

$$H.M = \frac{n}{\sum\left(\frac{f}{x}\right)} = \frac{48}{4.0283} = 11.9158$$

Results

1. if $G.M = A.M = H.M$ it is symmetrical
2. if $G.M \neq A.M \neq H.M$ it is skewed
3. if $A.M > G.M > H.M$ it is positive skewed
4. if $A.M < G.M < H.M$ it is negative skewed

In order to describe the data set more accurately, statisticians use measure of variations instead of measure of central tendency.

- Measure of central tendency

1. Mean
2. Median
3. Mode
4. Midrange
5. Weighted Mean

- Measure of variations

1. Range
2. Variance
3. Standard deviations

Measures of Variation: Range

- The range is the difference between the highest and lowest values in a data set, and is simplest of the three measures.

R = Highest – Lowest

Example: 20, 35, 67, 90, 75, 10

R = 90 – 10

R = 80

Ungrouped data

$$\text{Variance} = S^2 = \frac{\sum(x-\bar{x})^2}{n-1}$$

$$\text{Standard deviation} = S = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$$

Calculate Coefficient of Variation (C.V):

$$C.V = \frac{S.D}{\bar{x}} \times 100$$

$$\bar{x} = \frac{\sum x_i}{n}$$

Grouped data

$$\text{Variance} = S = \frac{\sum f(x-\bar{x})^2}{\sum f - 1}$$

$$\text{Standard deviation} = S = \sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f - 1}}$$

$$Mean = \bar{x} = \frac{\sum fx}{\sum f}$$

- To see the more meaningful statistic to measure the variability, statisticians use **Variance and Standard Deviations**.

- Population Variance and Standard Deviations

- Sample Variance and Standard Deviations

How to find the Population Variance and Population Standard Deviation.

1. **Find the mean for the data.** $\mu = \frac{\Sigma X}{N}$

2. **Find the Deviation for each data value.** $X - \mu$

3. **Square each of the deviations.** $(X - \mu)^2$

4. **Find the sum of the squares.** $\Sigma(X - \mu)^2$

5. **Divide by N to find the variance** $\sigma^2 = \frac{\Sigma(x-\mu)^2}{N}$

6. **Take the square root of the variance to find the standard deviation.**

$\sigma = \sqrt{\Sigma(x-\mu)^2/N}$

Find the variance and standard deviation for the data set for Brand A light bulb.

- 20, 34, 30, 35, 45, 40, 55

1. Find the mean. 20 + 34 + 30 + 35 + 45 + 40 + 55/7

$\mu = \frac{\Sigma x}{N}$ =259/7

=**37 weeks**

2. Subtract the mean from each data value. $x - \mu$

20 – 37 = -17   34 – 37 = -3   30 – 37 = -7   35 – 37 = -2

45 – 37 = 8   40 – 37 =3   55 – 37 = 18

3.Square each $(X-\mu)^2$

$(-17)^2 = 289$ $(-3)^2 = 9$ $(-7)^2 = 49$ $(-2)^2 = 4$

$(-8)^2 = 64$ $(-3)^2 = 9$ $(-18)^2 = 324$

4. Find the sum of the squares. $\Sigma(x-\mu)^2$

289 + 9 + 49 + 4 + 64 + 9 + 324 = 748

5.Divide the sum by N to find out the variance.

=748/7

= 106.857

6.Take the square root of the variance to get the standard deviation.

$\sqrt{106.857} \simeq 10.38$

Find the variance and standard deviation for the data set for Brand B light bulb.

- 10, 20, 30, 40, 50, 49, 60

1. Find the mean. 10 + 20 + 30 + 40 + 50 + 49 + 60/7

$\mu = \frac{\Sigma x}{N}$ =259/7

=**37 weeks**

2. Subtract the mean from each data value. $x - \mu$

10 – 37 = -27 20 – 37 = -17 30 – 37 = -7 40 – 37 = 3

50 – 37 = 13 49 – 37 = 12 60 – 37 = 23

Formula for sample Variance

- Formula

- X = Individual value
- $\overline{X}$ = sample mean
- n = sample size

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$\text{Standard deviation} = \sqrt{\text{Variance}}$$

How to find the Sample Variance and Sample Standar Deviation.

1. **Find the mean for the data.** $\bar{X} = \frac{\Sigma X}{n}$

2. **Find the Deviation for each data value.** $X - \bar{X}$

3. **Square each of the deviations.** $(X - \bar{X})^2$

4. **Find the sum of the squares.** $\Sigma (X - \bar{X})^2$

5. **Divide by n -1 to find the variance** $S^2 = \frac{\Sigma (X - \bar{X})^2}{n-1}$

6. **Take the square root of the variance to find the standard deviation.**
$S = \sqrt{\Sigma (X - \bar{X})^2 / n-1}$

Example: A sample of 6 taxi drivers shows the time they spent in school zone in rush hours. Find the Variance and standard deviation of the sample. 15,22,30,10,12,25

$$\bar{X} = \frac{\Sigma X}{n}$$

$$= \frac{15+22+30+10+12+25}{6}$$

$$= \frac{114}{6}$$

$$= 19 \text{ min}$$

2. Find the Deviation for each data value.

$X - \bar{X}$

15 - 19 = -4 22 - 19 = 3 30 -19 = 11 10 - 19 = -9

12 - 19 = -7 25 – 19 = 6

$(x-\bar{x})^2$

3. Square each of the deviations.

$(-4)^2 + (3)^2 + (11)^2 + (-9)^2 + (-7)^2 + (6)^2$

4. Find the sum of the squares.

$\sum (x-\bar{x})^2$

= 16 +9+121+81+49+36

= 312

5. Divide by n -1 to find the variance.

$$= \frac{312}{6-1}$$

$$S^2 = \frac{\Sigma (x-\bar{x})^2}{n-1}$$

= 312 /5

= 62.4 Variance

6. Take the square root of the variance to find the standard deviation.

$$S = \sqrt{\frac{\Sigma (x-\bar{x})^2}{n-1}}$$

$$= \sqrt{62.4}$$

$$\approx 7.9$$

Another method

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

sample variance 17 15 23 7 9 13

$$Mean = \bar{x} = \frac{\sum x}{n}$$

$$\bar{x} = \frac{17 + 15 + 23 + 7 + 9 + 13}{6} = 14$$

sample variance 17 15 23 7 9 13

x	$\bar{x}$	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$
17	14	3	9
15	14	1	1
23	14	9	81
7	14	-7	49
9	14	-5	25
13	14	-1	1
84		0	166

$\sum (x_i - \bar{x})^2$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$s^2 = \frac{166}{5} = 33.2$$

Compute the variance of the test scores.

scores (x)	frequency (f)
22-24	5
19-21	6
16-18	7
13-15	8
10-12	4

$$\text{variance} = \frac{\sum f(X-\bar{x})^2}{\sum f - 1}$$

scores (x)	frequency (f)	class mark (X)	fX	$\bar{x}$	$X-\bar{x}$	$(X-\bar{x})^2$	$f(X-\bar{x})^2$

scores (x)	frequency (f)	class mark (X)	fX	$\bar{x}$	$X-\bar{x}$	$(X-\bar{x})^2$	$f(X-\bar{x})^2$
22-24	5	23	115	17	6	36	180
19-21	6	20	120	17	3	9	54
16-18	7	17	119	17	0	0	0
13-15	8	14	112	17	-3	9	72
10-12	4	11	44	17	-6	36	144

$\sum f = 30$ $\sum fX = 510$

$\sum f - 1 = 30 - 1 = 29$ mean: $\bar{x} = 510/30 = 17$

$$\text{variance} = \frac{\sum f(X-\bar{x})^2}{\sum f - 1} = \frac{450}{30-1} = 15.51$$

Exercises :-1.1

1.1 The following measurements were recorded for the drying time, in hours, of a certain brand of latex paint. 3.4 2.5 4.8 2.9 3.6 2.8 3.3 5.6 3.7 2.8 4.4 4.0 5.2 3.0 4.8

Assume that the measurements are a simple random sample.

(a) What is the sample size for the above sample?

(b) Calculate the sample mean for these data.

(c) Calculate the sample median.

(d) Plot the data by way of a dot plot.

(e) Compute the 20% trimmed mean for the above data set.

(f) Is the sample mean for these data more or less descriptive as a center of location than the trimmed mean?

a) Number of measurements is recorded for the drying time gives the sample size n=15

b) b)

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$= \frac{1}{15}\left[3.4 + 2.5 + 4.8 + 2.9 + \ldots + 5.2 + 3.0 + 4.8\right]$$

$$= \frac{56.8}{15}$$

$$= 3.7867$$

c) We have to find the median for the given data.

Rearranged data in increasing order is

2.5,2.8,2.8,2.9,3,3.3,3.4,3.6,3.7,4,4.4,4.8,4.8,5.2 & 5.6 .

Then sample median = $x_{\frac{n+1}{2}}$ $as\ n\ is\ odd$

$$\Rightarrow \tilde{x} = x_8 = 3.6$$

d)

Dot plot for the given data is shown below.